The 22 Constructions

Junior & Leaving Certificate Mathematics

Instruments

Straight-edge draws a line through two points.

Compass draws arcs and circles, and transfers lengths.

Ruler measures and marks lengths.

Protractor measures and constructs angles.

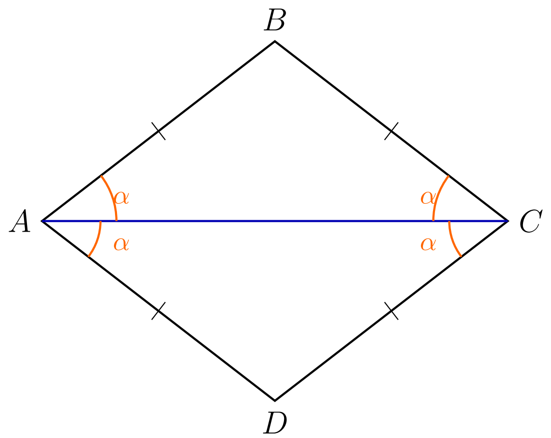
Set-squares construct right angles and standard angles (30°, 45°, 60°).

Constructions by syllabus level

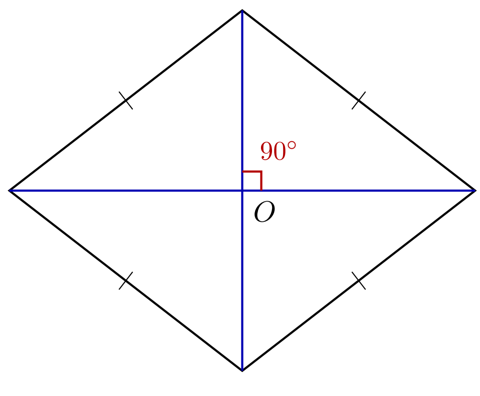
JC OL	1, 2, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15
JC HL	all of JC OL, plus **3** and **7**
LC OL	JC OL assumed, plus **16, 17, 18, 19, 20, 21**
LC HL	JC HL + LC OL, plus **22**
LC FL	revisits 4, 5, 10, 13, 15 in real-life contexts

Constructions

1. Bisector of an angle
2. Perpendicular bisector
3. **Perpendicular through external point**
4. Perpendicular through point on line
5. Parallel line through a point
6. Divide into 2 or 3 equal parts
7. **Divide into n equal parts**
8. Segment of given length on a ray
9. Angle of given size
10. Triangle from SSS
11. Triangle from SAS
12. Triangle from ASA
13. Right-angled triangle (hyp + side)
14. Right-angled triangle (side + angle)
15. Rectangle from side lengths
16. Circumcentre and circumcircle
17. Incentre and incircle
18. Angle of 60° without protractor
19. Tangent to circle at a point
20. Parallelogram from sides + angle
21. Centroid of a triangle
22. **Orthocentre of a triangle**



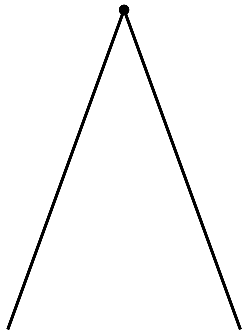
a)



b)

1. Bisector of an angle. The diagonals of a rhombus bisect its end angles.

a)



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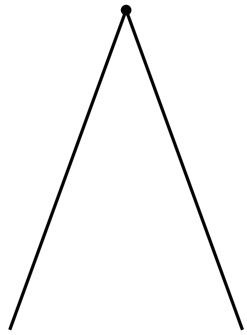

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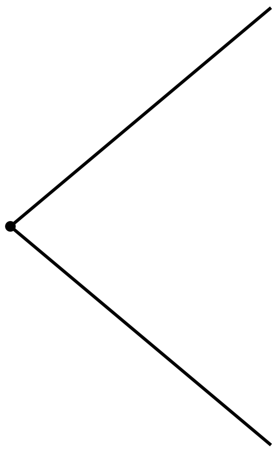

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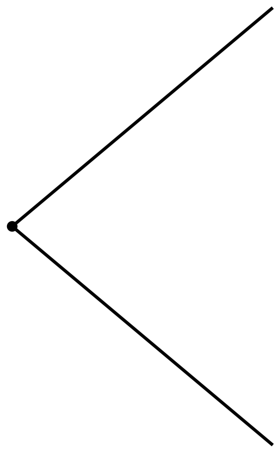

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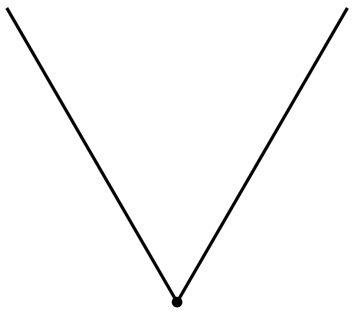

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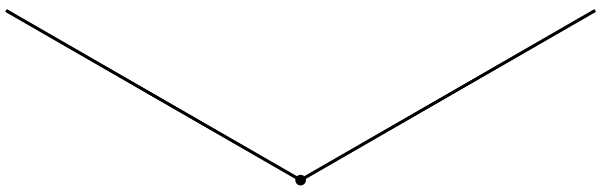


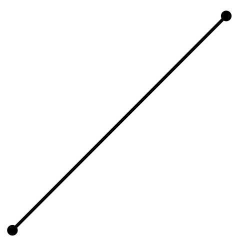
2. Construct the perpendicular bisector of each line segment, using only a compass and straight edge.

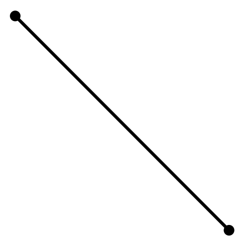
a)

b)

c) 

d) 

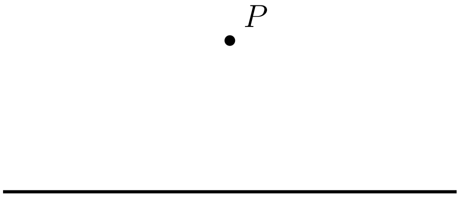
e) 

f) 

3. Perpendicular through a point not on the line. *(JC HL)* Construct the perpendicular to each line, passing through the point P which is not on the line.

   from P, swing an arc cutting the line at two points, then bisect the segment between them.

a)



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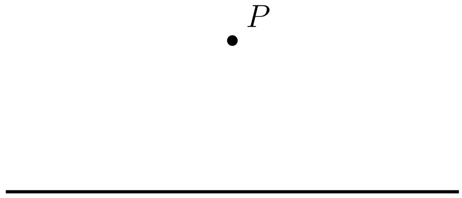

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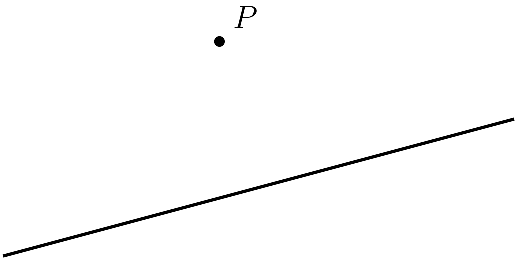

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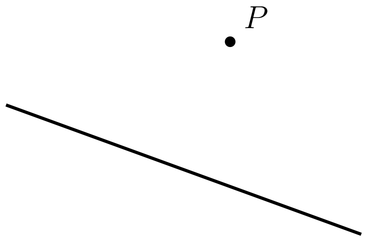

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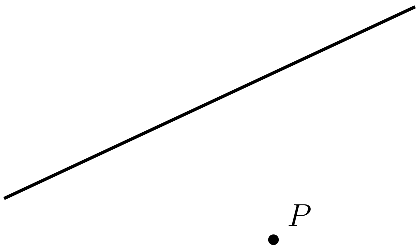

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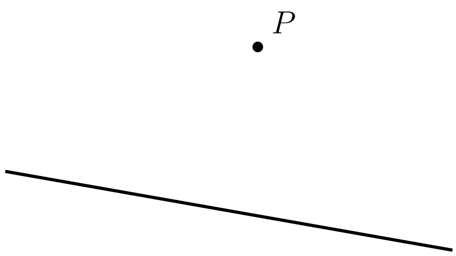
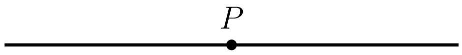


4. Perpendicular through a point on the line. Construct the perpendicular to each line through the point P on the line.

a) 

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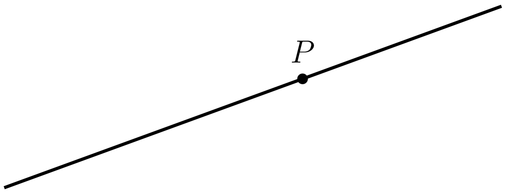
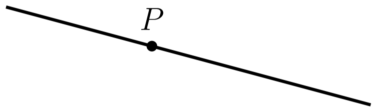
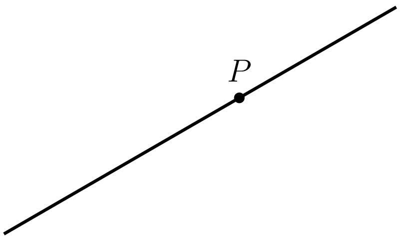
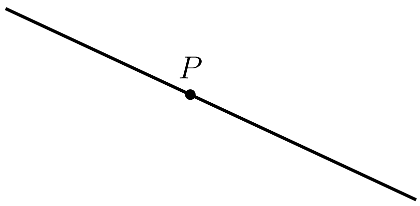
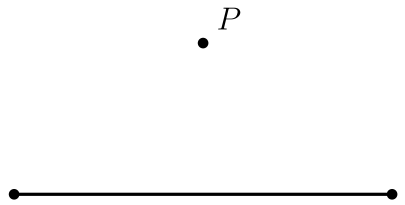
c) 

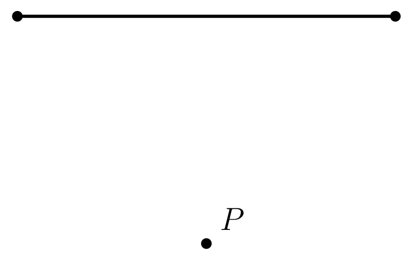
d) 

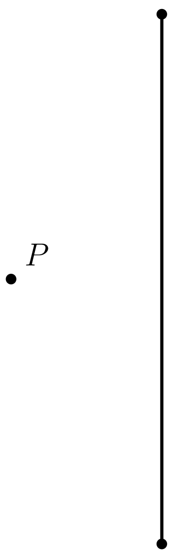
e) 

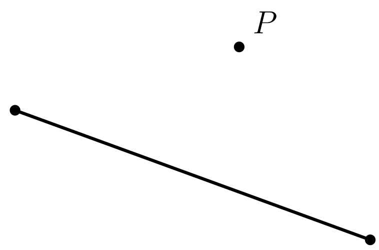
f) 

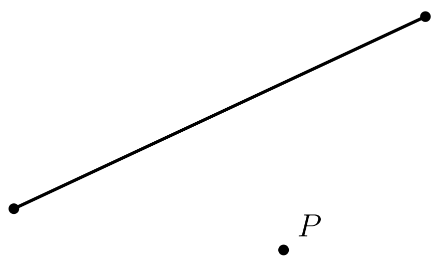
5. Parallel line through a given point. Construct the line through the given point P parallel to the given line.

a) 

b) 

c) 

d) 

e) 

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